RISK 201: Simple Annuities

Himchan Jeong, FSA

Department of Statistics & Actuarial Science
Simon Fraser University

3.1 Definitions

- An **annuity** is a series of payments made at regular intervals of time.
- The **payment period** is the interval between annuity payments.
- An **annuity-certain** is an annuity where the payments are sure to be made for a fixed period of time (the **term** of the annuity).
- A **contingent annuity** is an annuity where the payments depend on one or more contingent events.

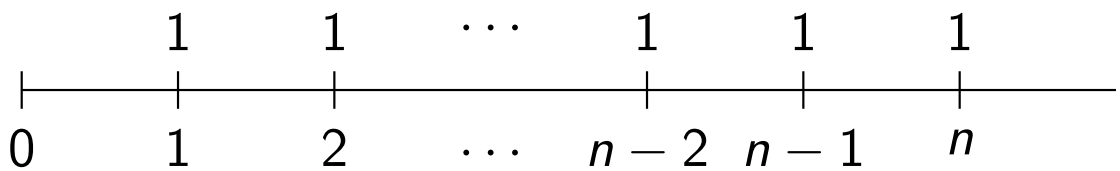
Types of annuities

Ordinary annuity	Payments made at the **end** of each payment period
Annuity due	Payments made at **beginning** of each payment period
Deferred annuity	First payment at a **later** date
Simple annuity	Payment interval and compounding period are the same (Ch 3)
General annuity	Payment interval and compounding period are different (Ch 4)

3.2 AV of an ordinary annuity

When all payments are 1 and the effective rate of interest is i, the accumulated value (at time n) of an annuity-immediate with n payments is denoted by

When there is no ambiguity about i, we use

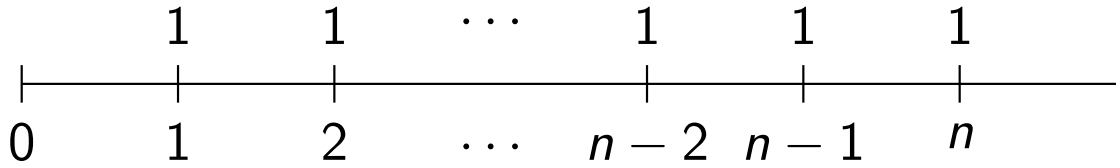


Sum of a geometric progression

Recall the sum of finite terms in a geometric progression:

$$S = a + ar + ar^2 + ar^3 + \cdots + ar^{n-1}$$
$$= \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1}$$

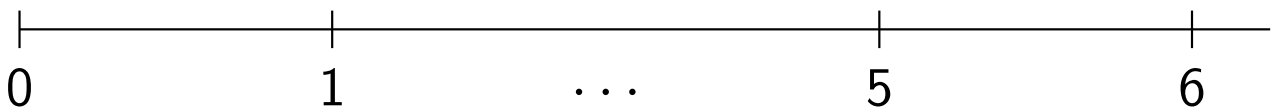
AV of an ordinary annuity, cont'd



$$
\begin{aligned}
s_{\overline{n}|i} &= 1 + (1+i) + (1+i)^2 + \cdots + (1+i)^{n-1} \\
&= \frac{(1+i)^n - 1}{i}
\end{aligned}
$$

Example 3.1

Emma is starting a 6-year university degree. Her grandmother wants to support her by giving her a \$5000 cash gift for each year she completes. Instead of spending this money, Emma decides to invest it. Assuming a 6% effective rate of interest, how much will Emma have in her investment account right after the last gift she receives?



$$\text{AV} = 5000\, s_{\overline{6}|0.06} = \mathbf{34{,}876.59}$$

Using a financial calculator

N = the number of compounding periods

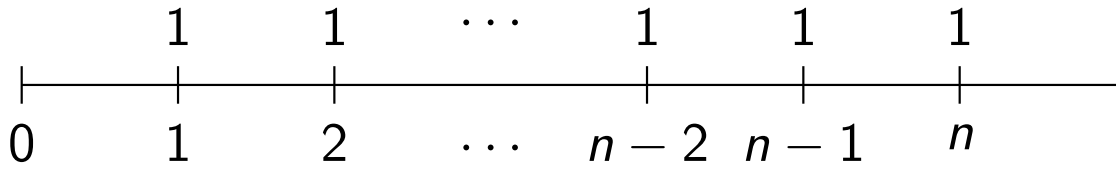
PV = 0

I/Y = interest rate per compounding period

PMT = periodic payments

CPT FV

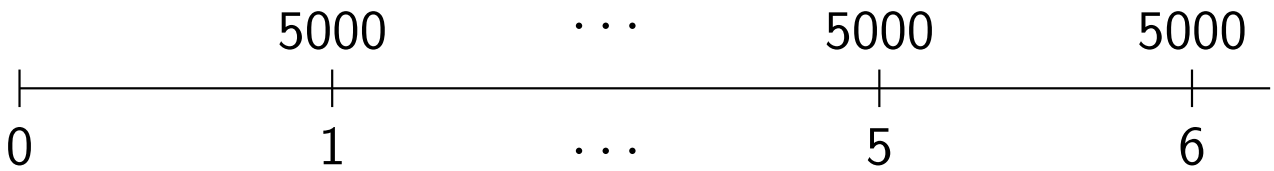
3.3 PV of an ordinary annuity



$$\begin{aligned} a_{\overline{n}|} &= v + v^2 + \cdots + v^{n-1} + v^n \\ &= \frac{1 - v^n}{i} \end{aligned}$$

Example 3.1, continued

If Emma's grandmother expects her to finish all 6 years of university successfully, how much should she set aside today to fund the gifts? Assume she too can earn a 6% effective rate of interest.



$\text{PV} = 5000\, a_{\overline{6}|0.06} = \mathbf{24{,}586.62}$

Using a financial calculator

N = the number of compounding periods

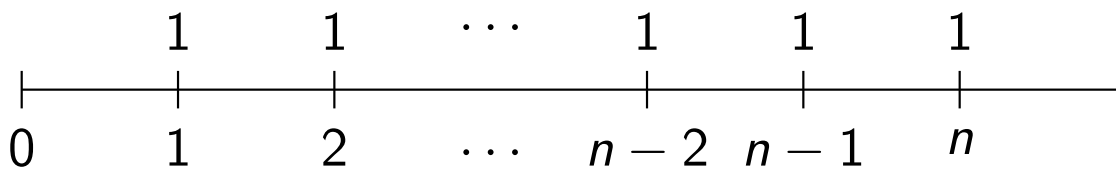
FV = 0

I/Y = interest rate per compounding period

PMT = periodic payments

CPT PV

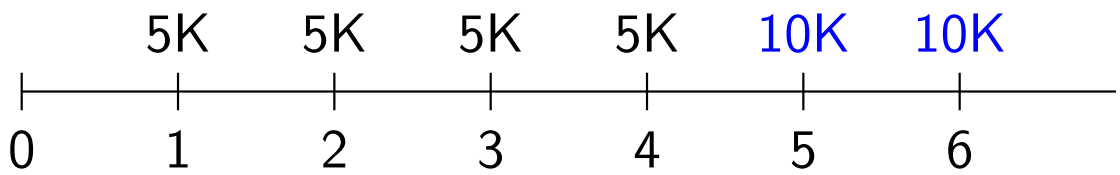
Relationship between $a_{\overline{n}|}$ and $s_{\overline{n}|}$



$$a_{\overline{n}|} = v^n \quad s_{\overline{n}|}$$

Changing payment amounts

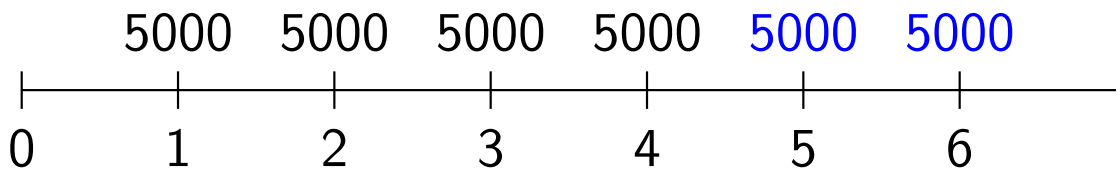
Example 3.2: Emma's grandmother decides that Emma should get \$5,000 in each of the first 4 years of her studies, and then \$10,000 in each of the last 2 years. How much should the grandmother set aside in an account today earning a 6% effective annual rate of interest?



$$\begin{aligned} \text{PV} &= 5000\, a_{\overline{6}|0.06} + 5000 v^5 + 5000 v^6 \\ &= 5000\, a_{\overline{4}|0.06} + v^4\, 10000\, a_{\overline{2}|0.06} \end{aligned}$$

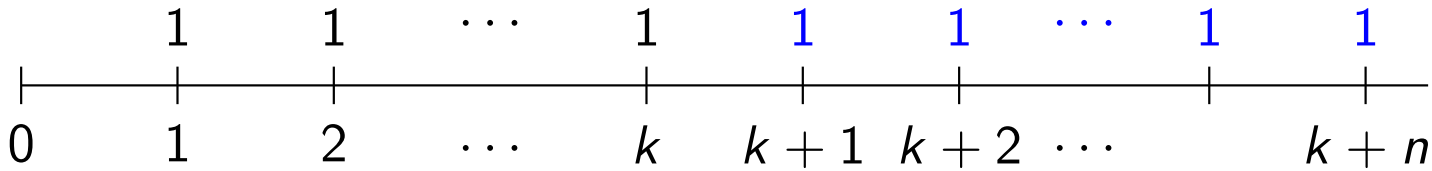
Changing interest rates

Example 3.3: Suppose Emma receives \$5000 a year and invests it. During her first four years of school she earns 6% effective interest and during her last 2 years she earns 5% effective. What is the acc'd value just after she receives the last payment?

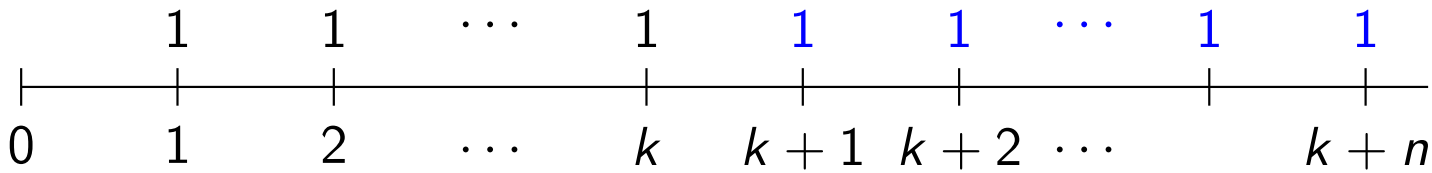


$$\text{AV} = 5000\, s_{\overline{2}|0.05} + 5000\, s_{\overline{4}|0.06}(1.05)^2$$

Splitting payment streams

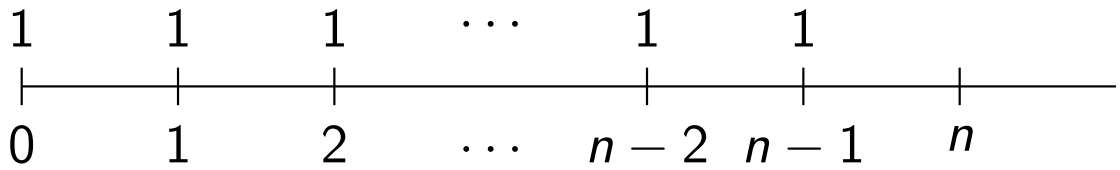


$$a_{\overline{k+n}|} = a_{\overline{k}|} + v^k a_{\overline{n}|}$$



$$s_{\overline{k+n}|} = (1+i)^n s_{\overline{k}|} + s_{\overline{n}|}$$

3.4 Annuities due

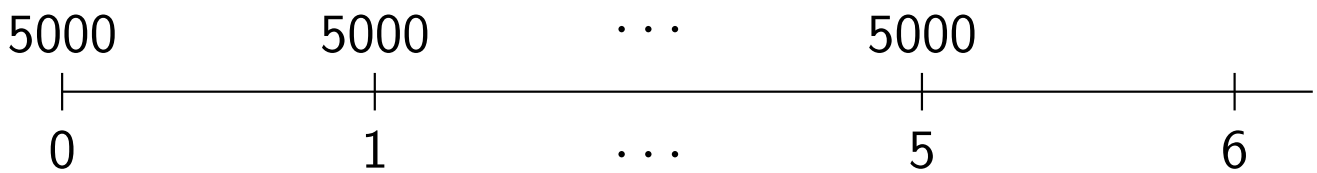


Accumulated value at the end of the term (n years):

$$\begin{aligned}\ddot{s}_{\overline{n}|} &= (1+i)+(1+i)^2+\cdots+(1+i)^n \\ &= \frac{(1+i)\big((1+i)^n-1\big)}{1+i-1} = \frac{(1+i)^n-1}{i/(1+i)} = \frac{(1+i)^n-1}{d}\end{aligned}$$

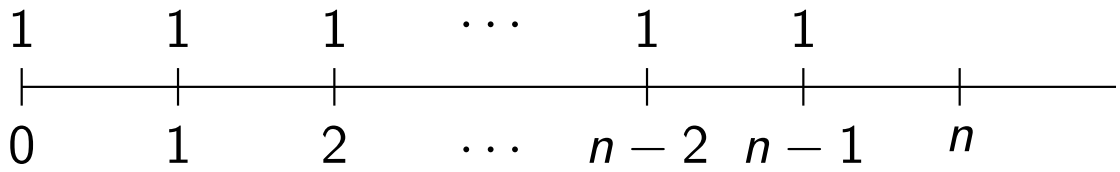
Example 3.1, continued

Suppose Emma's grandmother wants to gift her at the **beginning** of each year she attempts, rather than at the end of each year. How much would Emma have when she finishes her degree if she invests the gifts at a 6% effective rate of interest?



$$\text{AV} = 5000\,\ddot{s}_{\overline{6}|0.06} = (1.06)\,5000\,s_{\overline{6}|0.06} = \mathbf{36{,}969.19}$$

PV of annuity due

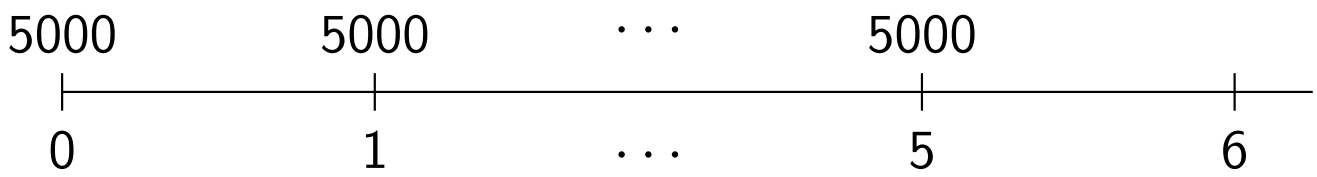


Present value at the beginning of the term:

$$\ddot{a}_{\overline{n}|} = 1 + v + v^2 + \cdots + v^{n-1}$$
$$= \frac{1 - v^n}{1 - v} = \frac{1 - v^n}{d}$$

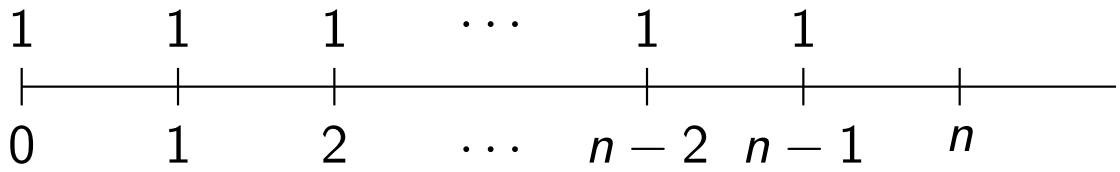
Example 3.1, continued

How much should Emma's grandmother set aside today to fund the beginning-of-year gifts, if she can earn a 6% effective rate of interest?



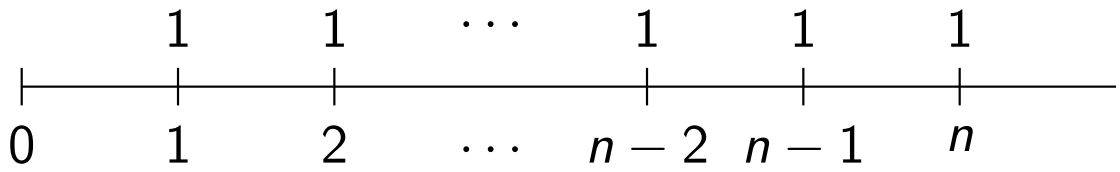
$$\text{PV} = 5000\,\ddot{a}_{\overline{6}|0.06} = (1.06)\,5000\,a_{\overline{6}|0.06} = \mathbf{26{,}061.82}$$

Relationship between $\ddot{a}_{\overline{n}|}$ and $\ddot{s}_{\overline{n}|}$



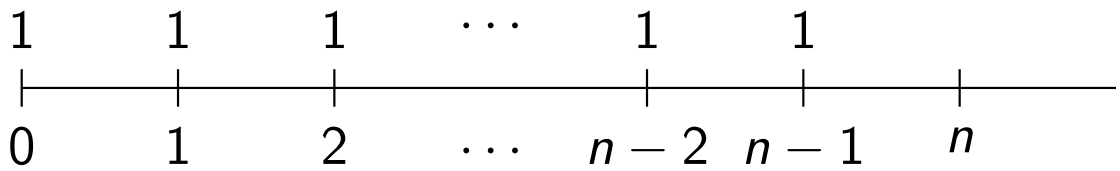
$$\ddot{a}_{\overline{n}|} = v^n \ddot{s}_{\overline{n}|}$$

More relationships



$$a_{\overline{n}|} = v\, \ddot{a}_{\overline{n}|}$$

More relationships, continued



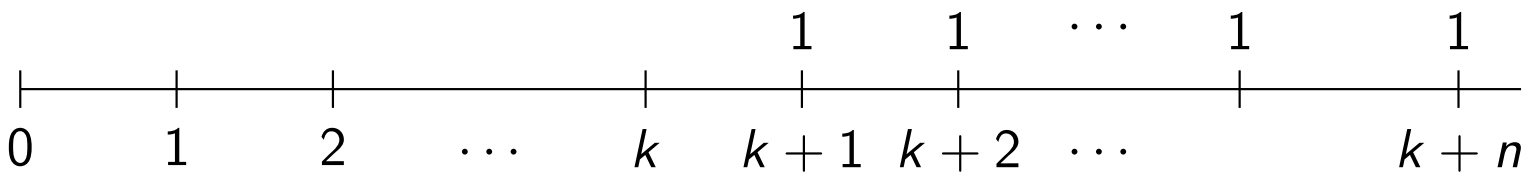
$$s_{\overline{n}|} = v\, \ddot{s}_{\overline{n}|}$$

Summary

Notation	# of pmts	Focal date	Formula
$a_{\overline{n}\vert}$	n	One period before 1st payment	$\frac{1-v^n}{i}$
$\ddot{a}_{\overline{n}\vert}$	n	At time of 1st payment	$\frac{1-v^n}{d}$
$s_{\overline{n}\vert}$	n	At time of last payment	$\frac{(1+i)^n-1}{i}$
$\ddot{s}_{\overline{n}\vert}$	n	One period after last payment	$\frac{(1+i)^n-1}{d}$

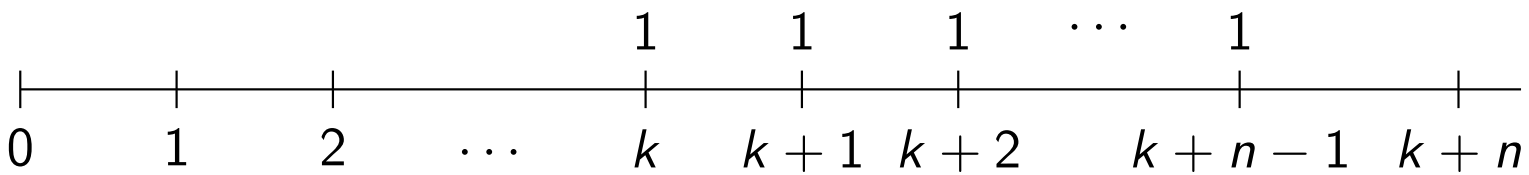
Deferred annuities

k-year deferred, n-year ordinary annuity:



$$\text{PV} = v^k \, a_{\overline{n}|}, \quad \text{AV} = s_{\overline{n}|}$$

k-year deferred, n-year annuity due:



$$\text{PV} = v^k \, \ddot{a}_{\overline{n}|}, \quad \text{AV} = \ddot{s}_{\overline{n}|}$$

Annuity recap

Example 3.4: Using $i = 6\%$, find the value of a 10-year annuity with payments of \$1 starting at time 2, determined at:

(a) $t = 0$: $v^2 \, \ddot{a}_{\overline{10}|0.06} = v \, a_{\overline{10}|0.06}$

(b) $t = 1$: $v \, \ddot{a}_{\overline{10}|0.06} = a_{\overline{10}|0.06}$

(c) $t = 2$: $\ddot{a}_{\overline{10}|0.06} = (1.06) \, a_{\overline{10}|0.06}$

(d) $t = 4$: $(1.06)^2 \, \ddot{a}_{\overline{10}|0.06}$

(e) $t = 12.5$: $(1.06)^{10.5} \, \ddot{a}_{\overline{10}|0.06}$

3.5 Finding the term of an annuity

Often you will need to find the term of an annuity, given the present value, the regular payments and an interest rate.

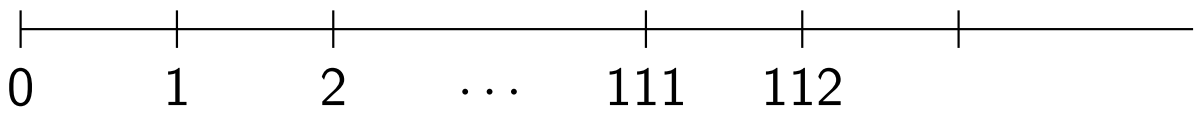
Most of the time, the resulting term will not be an integer multiple of the payment period.

Example 3.5: You borrow \$54,000 at $i^{(12)} = 3\%$ to buy a car. If the most you can afford to pay is \$550/month, how long will it take to pay off the loan?

Balloon payments

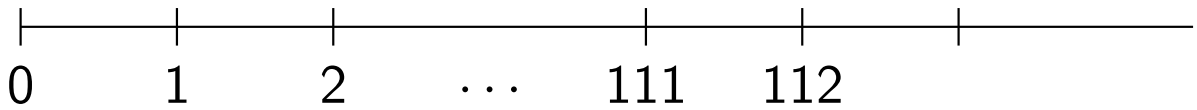
Instead of making a payment at an irregular time, we can adjust the last payment.

One option is to increase the last payment. This is called a balloon payment.



Drop payments

Another option is to add a smaller payment one period after the last regular payment. This is called a drop payment.



Common mistakes

Q: My calculator gives me a negative value for N. What do I do?

A: You probably did not enter the sign of the payment (PMT) correctly. If your PV is positive, PMT needs to be negative in order to repay the loan.

Q: I keep making calculation mistakes. How can I check the reasonableness of my answer?

A: The time to pay back a loan should be more than PV $\div$ Payments. Always draw a time diagram. To check your answer, try a different focal date.

3.6 Finding the interest rate

Sometimes you know the PV (loan amount) or FV (the desired savings at some future date), as well as the payments and the term, so what you need to find is the interest rate.

This is easy if you have a financial calculator.

Your calculator actually runs an approximation algorithm to find the interest rate that satisfies your equation of value.

You can do this too.

Approximation methods

1. Consider rates in 1% increments, use linear approximation between closest rates
2. Proper root-finding algorithms
    a. Bisection
    b. Newton–Raphson method
3. In Excel, you can use the RATE function or the Goal Seek feature (under Tools menu)
4. In R, the NLRoot package has both bisection and Newton-Raphson methods

Starting values

If you know the future value and the term, use

$$i = \frac{\left(\frac{FV}{n}\right)^2 - 1}{FV}$$

as your initial guess.

If you know the present value and the term, use

$$i = \frac{1 - \left(\frac{PV}{n}\right)^2}{PV}.$$

Example 3.6

A 50,000 loan is being repaid with payments of 1200 at the end of each month for 5 years. Find $i^{(12)}$ being charged on the loan.

$$\frac{i^{(12)}}{12} = \mathbf{1.283\%}, \qquad i^{(12)} = 15.396\%$$

Example 3.7

A 50,000 loan is being repaid with payments of 1200 at the end of each month for first 2 years, and then 1500 for the last 3 years. Find $i^{(12)}$ being charged on the loan.

$$\begin{aligned} 50000 &= 1200\, a_{\overline{24}|j} + \frac{1}{(1+j)^{24}}\, 1500\, a_{\overline{36}|j} \\ &= 1200\, \frac{1-v^{24}}{j} + v^{24}\, 1500\, \frac{1-v^{36}}{j} \end{aligned}$$

$$50000 - 1200\, \frac{1-v^{24}}{j} - v^{24}\, 1500\, \frac{1-v^{36}}{j} = 0$$